

## Signal Processing

### Exercise Sheet 11 – FIR Filter

SoSe 2026

#### Question 1

An FIR low-pass filter is required to have the following specifications:

$$0.95 < |H(e^{j\omega})| < 1.05 \quad 0 \leq |\omega| < 0.25\pi$$

$$|H(e^{j\omega})| < 0.1 \quad 0.35\pi \leq |\omega| < \pi$$

Draw the tolerance scheme and determine which window functions are suitable to meet the above listed specifications. Assume that we would like to design the low-pass filter using the window method. Calculate the minimum order  $N$  of the low-pass filter for each suitable window function.

#### Question 2

An FIR low-pass filter is required to have the following specifications:

- Pass-band edge frequency  $f_{\text{pass}} = 2$  kHz.
- Transition width  $\Delta f = 200$  Hz.
- Pass-band ripple  $\delta_{\text{pass}} = 0.0116$ .
- Minimum stop-band attenuation  $\delta_{\text{stop}} = 50$  dB.
- Sampling frequency  $f_s = 10$  kHz.

We wish to design the digital filter using the window method. Determine an appropriate window function and calculate the required filter order to design the filter. Furthermore, derive the equation to determine the filter coefficients.

### Question 3

By using the window method, a discrete-time filter is to be designed. The filter, which should meet the following specifications, is to be designed with a Kaiser window.

$$|H(e^{j\omega})| \leq 0.01 \quad 0 \leq |\omega| < 0.25\pi$$

$$0.95 \leq |H(e^{j\omega})| \leq 1.05 \quad 0.35\pi \leq |\omega| < 0.6\pi$$

$$|H(e^{j\omega})| \leq 0.01 \quad 0.65\pi \leq |\omega| < \pi$$

- Determine the minimum length  $N + 1$  of the impulse response and the value of the Kaiser windows parameter  $\beta$ .
- Calculate the group delay.
- Determine the ideal impulse response  $h_d[n]$  to which the Kaiser window should be applied to.

### Question 4

The impulse response of an FIR filter is given as:  $h[n] = [0.0565, 0.4435, 0.4435, 0.0565]$ .

- Determine the difference equation of this filter system.
- Write down the expression of  $H(e^{j\omega})$  and calculate the magnitude response  $|H(e^{j\omega})|$  at  $\omega = 0, 0.5\pi, \pi$ .
- Is this filter a low-pass, band-pass or high-pass filter?
- The filter was designed using a rectangular window. Determine the maximum error in the passband and in the stopband. State the result linearly and in dB.
- Calculate the group delay of the filter.

## FIR Filter Design – Cheat Sheet

### Window Method

| Window Type | Function                                                                                       | Peak Sidelobe Amplitude (Relative) in dB | Approximate Width of Main Lobe | Peak Approximation Error in dB |
|-------------|------------------------------------------------------------------------------------------------|------------------------------------------|--------------------------------|--------------------------------|
| Rectangular | $w(n) = 1$                                                                                     | -13                                      | $\frac{4\pi}{N+1}$             | -21                            |
| Bartlett    | $w(n) = 1 - \frac{2 n - \frac{N}{2} }{N}$                                                      | -25                                      | $\frac{8\pi}{N}$               | -25                            |
| Hann        | $w(n) = 0.5 - 0.5 \cos\left(\frac{2\pi n}{N}\right)$                                           | -31                                      | $\frac{8\pi}{N}$               | -44                            |
| Hamming     | $w(n) = 0.54 - 0.46 \cos\left(\frac{2\pi n}{N}\right)$                                         | -41                                      | $\frac{8\pi}{N}$               | -53                            |
| Blackman    | $w(n) = 0.42 - 0.5 \cos\left(\frac{2\pi n}{N}\right) + 0.08 \cos\left(\frac{4\pi n}{N}\right)$ | -57                                      | $\frac{12\pi}{N}$              | -74                            |

| Window Type | Function                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Kaiser      | $w(n) = \frac{I_0\left[\beta \sqrt{1 - \left(\frac{2n}{N-1}\right)^2}\right]}{I_0(\beta)}$ $I_0(x) = \sum_{k=0}^{\infty} \left(\frac{x^k}{2^k k!}\right)^2$ $A = -20 \log(\delta)$ $\Delta\omega = \omega_{\text{stop}} - \omega_{\text{pass}}$ $\beta = \begin{cases} 0.1102(A - 8.7) & A > 50 \\ 0.5842(A - 21)^{0.4} + 0.07886(A - 21) & 21 \leq A \leq 50 \\ 0.0 & A < 21 \end{cases}$ $N = \frac{A - 8}{2.285\Delta\omega}$ |

### Optimal Method

Minimum order  $N_e$  of the filter for the optimal method:

$$N_e = \frac{2}{3} \log\left(\frac{1}{10\delta_{\text{pass}}\delta_{\text{stop}}}\right) \cdot \frac{\omega_{\text{stop}}}{\omega_{\text{stop}} - \omega_{\text{pass}}}.$$